

2024 Higher Chemistry Paper 2 - Q5

Section: Nature's Chemistry

Topic: Alcohols, Fats and Oils

Question summary (Q5):

Fats and oils are important dietary components.

(a) State a reason why fats and oils form part of a balanced diet.

(b) Fats and oils are formed using the alcohol glycerol.

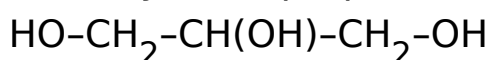
(i) Draw the structural formula for glycerol.

(ii) Name the functional group present in all fats and oils.

Worked Solution:

(a) Fats and oils provide a concentrated source of energy and are also needed for the transport/absorption of fat-soluble vitamins (A, D, E, K).

(b) (i) Glycerol (propane-1,2,3-triol) has the structural formula:



(b) (ii) The functional group in all fats and oils is the ester group ($-\text{COO}-$).

Final Answer:

(a) Provide energy / absorb fat-soluble vitamins.

(b) (i) $\text{HO}-\text{CH}_2-\text{CH}(\text{OH})-\text{CH}_2-\text{OH}$

(b) (ii) Ester group.

Revision Tips:

- Glycerol = propane-1,2,3-triol: a tri-alcohol.
- Fats and oils are esters formed from glycerol + fatty acids.
- Functions of fats and oils: energy storage, insulation, vitamin transport.